



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY

9189/3

PAPER 3 Multiple Choice

JUNE 2017 SESSION

1 hour

Additional materials:

Data Booklet

Mathematical tables and/or calculator

Multiple Choice answer sheet

Soft clean eraser

Soft pencil (type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**. Choose the one you consider correct and record your choice in soft pencil on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score **one** mark. A mark will **not** be deducted for a wrong answer. Any rough working should be done in this booklet.

This question paper consists of 13 printed pages and 3 blank pages.

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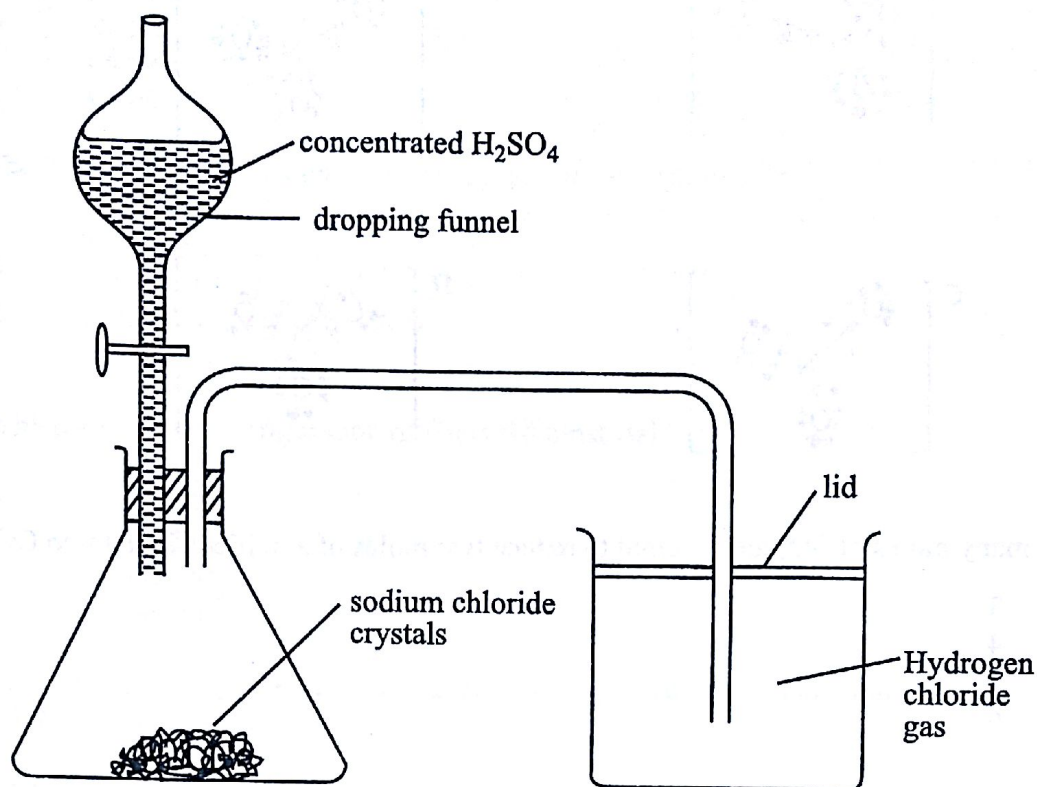
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Section A

For each question there are four possible answers, A, B, C and D. Choose the one you consider to be correct.

- 1 Which equation shows the first ionization energy of oxygen?
- A $\text{O}_{2(g)} \rightarrow 2\text{O}_{(g)}^+ + \text{e}^-$
- B $\text{O}_{2(g)} \rightarrow \text{O}_{2(g)}^+ + \text{e}^-$
- C $2\text{O}_{(g)} \rightarrow 2\text{O}_{(g)}^+ + 2\text{e}^-$
- D $\text{O}_{(g)} \rightarrow \text{O}_{(g)}^+ + \text{e}^-$
- 2 Which statement best explains why the first ionisation energy of magnesium, Mg and aluminium, Al differs?
- A lower shielding effect in Al
- B 3 p-orbital of Al has a lower energy level
- C the s-orbital of Mg is full
- D higher effective nuclear attraction in Al
- 3 Which substance has a regular tetrahedral structure?
- A silicon
- B diamond
- C silicon dioxide
- D silicon carbide

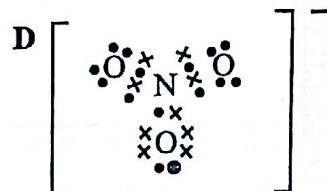
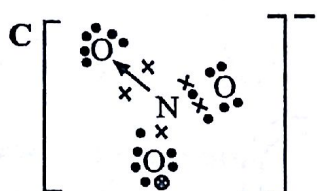
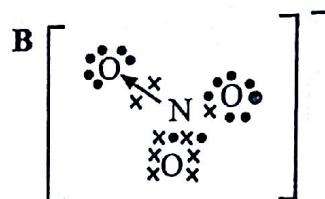
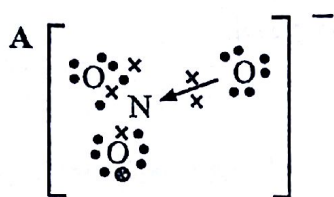
- 4 The diagram shows how hydrogen chloride can be prepared.



The method of collecting $\text{HCl}_{(\text{g})}$ is

- A upward displacement because $\text{HCl}_{(\text{g})}$ is denser than air.
 - B downward displacement because $\text{HCl}_{(\text{g})}$ is denser than air.
 - C upward delivery because $\text{HCl}_{(\text{g})}$ is less dense than air.
 - D downward delivery because $\text{HCl}_{(\text{g})}$ is less dense than air.
- 5 The unreactivity of nitrogen may be attributed to the nitrogen triple bond which is
- A short and difficult to break.
 - B long and easy to break.
 - C long and difficult to break.
 - D short and easy to break.

6 Which dot-and-cross diagram of the nitrate ion, NO_3^- , is correct?



7 How many moles of SO_2 are required to reduce two moles of acidified $\text{K}_2\text{Cr}_2\text{O}_7$ to Cr^{3+} ?

- A 3
- B 4
- C 5
- D 6

- 8 Which type of particles are present in the iodine crystal?
- A molecules
 - B cations
 - C anions
 - D atoms
- 9 The incomplete combustion in a car engine can be explained by
- A lack of oxygen.
 - B high temperatures in the engine.
 - C lack of catalytic converters.
 - D short reaction time.
- 10 Which electronic configuration of Cu is the most stable?
- A $[\text{Ar}] 3d^9 4s^2$
 - B $[\text{Ar}] 4s^2 3d^9$
 - C $[\text{Ar}] 3d^{10} 4s^1$
 - D $[\text{Ar}] 4s^1 3d^{10}$
- 11 Temperature changes were investigated when equal volumes of any two of the following solutions were mixed.
- FA4: 1 mol dm⁻³ NaOH_(aq)
 FA5: 1 mol dm⁻³ HCl_(aq)
 FA6: 2 mol dm⁻³ HCl_(aq)

Which observation is correct for the given mixture?

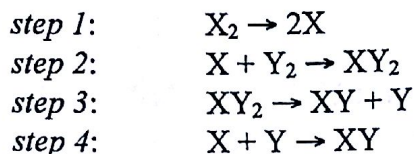
	mixture	observation
A	FA4 + FA6	temperature rise > temperature rise when FA4 + FA5 are mixed
B	FA4 + FA5	temperature rise = temperature rise when FA4 + FA6 are mixed
C	FA4 + FA5	temperature rise > temperature rise when FA4 + FA6 are mixed
D	FA5 + FA6	temperature rise > temperature rise when FA4 and FA5 are mixed

- 12 In an experiment 0.6 moles of sulphur dioxide were mixed with 2 moles of oxygen at 1.2 atmospheres and 400 °C. At equilibrium 0.4 moles of sulphur trioxide were present.

How many moles of oxygen were also present at equilibrium?

- A 1.8 moles
- B 1.6 moles
- C 0.4 moles
- D 0.2 moles

- 13 A proposed reaction mechanism for a certain reaction is:



Which statement is true about the reaction?

- A The overall order of reaction is 3.
- B The rate depends on the concentration of Y_2 .
- C The overall reaction equation is $\text{X} + \text{Y}_2 \rightarrow \text{XY}_2$
- D The overall reaction equation is $\text{X}_2 + \text{Y}_2 \rightarrow 2\text{XY}$

- 14 Chlorine gas is used for purification of drinking water.

How are the pathogens killed?

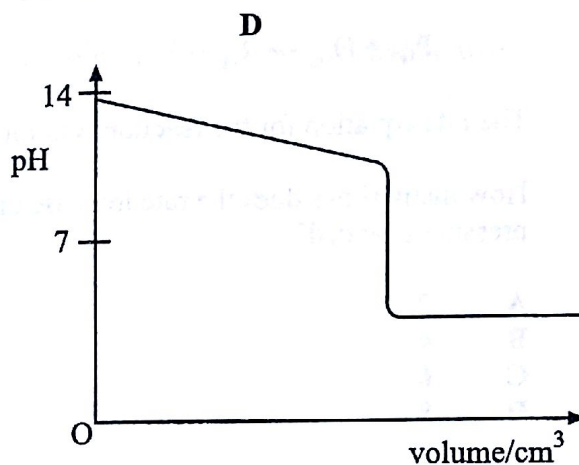
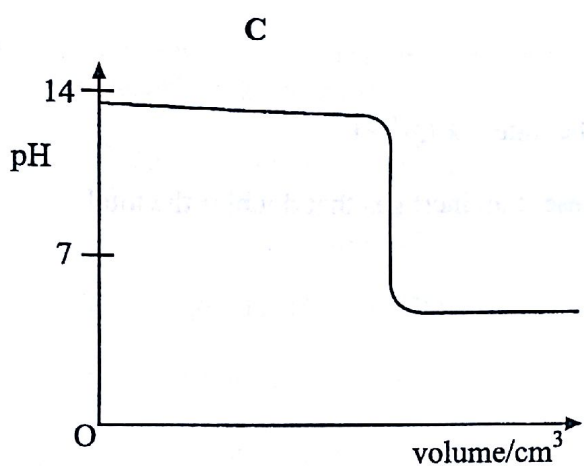
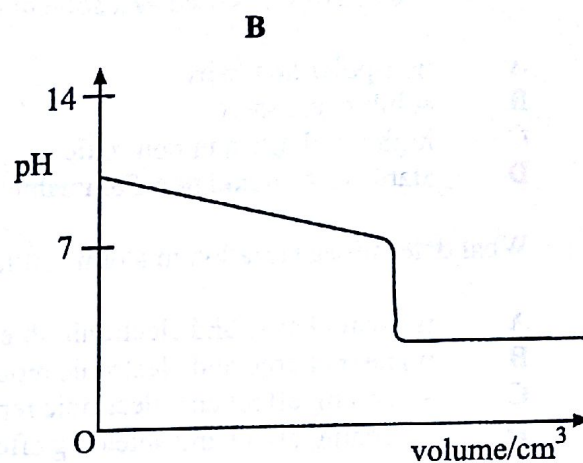
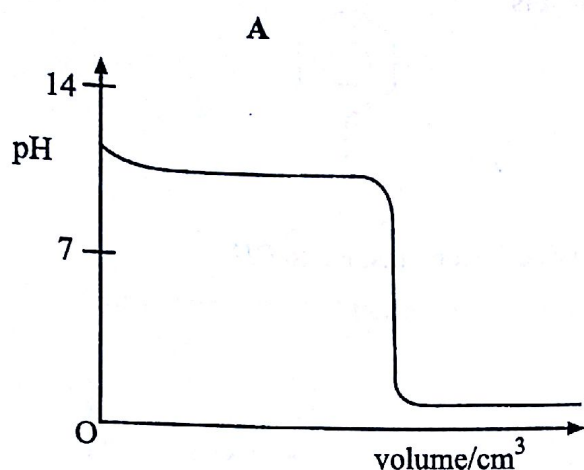
- A The strong acid, HCl , corrodes the bacteria.
- B The acid, HClO , oxidises the bacteria.
- C The smell of chlorine kills the bacteria.
- D The chlorine gas bleaches the bacteria.

- 15 Volumes of 25 cm³ of 2 moldm⁻³ $\text{NaOH}_{(\text{aq})}$, 25 cm³ of 1 moldm⁻³ $\text{NaOH}_{(\text{aq})}$ and 25 cm³ of 0.5 moldm⁻³ $\text{NaOH}_{(\text{aq})}$ were mixed thoroughly. 25 cm³ of 0.25 moldm⁻³ $\text{HCl}_{(\text{aq})}$ was added to the mixture.

The resultant concentration of $\text{NaOH}_{(\text{aq})}$ was

- A 3.2500 moldm⁻³.
- B 0.8750 moldm⁻³.
- C 0.8125 moldm⁻³.
- D 0.7500 moldm⁻³.

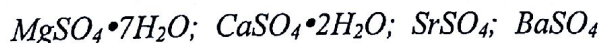
- 16 Which diagram correctly shows the pH-changes during the titration of ammonia solution against hydrochloric acid?



- 17 Which statement is **not** true about a buffer solution?

- A Its pH does not change on addition of little amounts of a weak acid.
- B Its pH does not change on addition of a weak base.
- C Its pH changes on addition of little amounts of strong acid.
- D Its pH does not change when the acid: salt ratio is changed.

- 18 The crystalline form of Group (II) sulphates may be hydrated or anhydrous as:



What is the best explanation for this trend of hydration?

- A The ionic character of the sulphates decreases down the group.
- B The charge density of the sulphates decreases down the group.
- C Lattice Energy of the sulphates becomes more exothermic down the group.
- D Hydration energy of the sulphates becomes more exothermic down the group.

- 19 Tin (IV) iodide is prepared by mixing powdered tin with iodine dissolved in carbon tetrachloride.

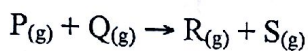
Carbon tetrachloride is used as a solvent because it is

- A non-polar and inert.
- B soluble and inert.
- C highly volatile and non-toxic.
- D stable to heat and non-flammable.

- 20 What determines variation in atomic radius of Period 3 elements, Na to Cl?

- A nuclear charge and electronic shielding
- B nuclear charge and electronic repulsion
- C screening effect and electronic repulsion
- D screening effect and shielding effect

- 21 Gases P and Q react according to the equation:



The rate equation for the reaction was found to be: $\text{rate} = k[Q]^2[P]$.

How many times does the rate increase or decrease if an inert gas that doubles the total pressure is added?

- A 2
- B 4
- C 6
- D 8

- 22 A solution of $Mn^{2+}_{(aq)}$, $Al^{3+}_{(aq)}$ and $Zn^{2+}_{(aq)}$ was treated with excess dilute NaOH.

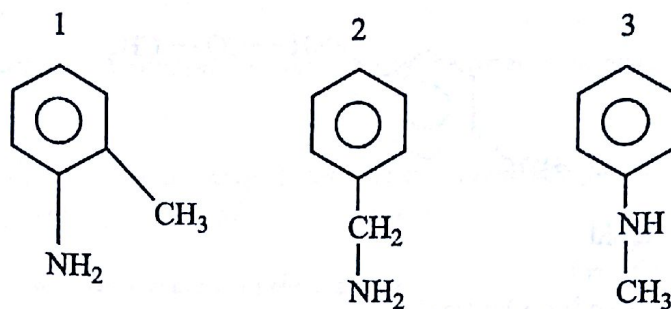
Which precipitate(s) are formed?

- A $Al(OH)_{2(s)}$ and $Mn(OH)_{2(s)}$
- B $Zn(OH)_{2(s)}$ and $Mn(OH)_{2(s)}$
- C $Zn(OH)_{2(s)}$ only
- D $Mn(OH)_{2(s)}$ only

- 23 Which word best describes the catalytic production of ethene from ethanol?

- A cracking
- B dehydrogenation
- C dehydration
- D hydrolysis

- 24 Three structures of organic compounds are shown.

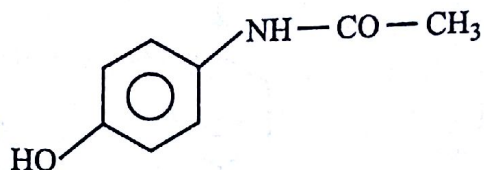


The correct order of increasing basic strength of these compounds is

- A 1 → 2 → 3.
 B 1 → 3 → 2.
 C 2 → 1 → 3.
 D 3 → 1 → 2.
- 25 A compound, E, decolourises bromine water and gives an orange red precipitate with 2,4-dinitrophenylhydrazine.
- Which is the possible structure of E?
- A $\text{CH}_3\text{CH}_2\text{CH}_2\text{COCN}$
 B $\text{O}=\text{CHCH}_2\text{CH}_2\text{CHO}$
 C $\text{CH}_2\text{CHCH}_2\text{CHO}$
 D $\text{CH}_2\text{CHCH}_2\text{CH}_2\text{OH}$
- 26 Which compounds react to give an observable change?

- A alkaline I_2 + $\text{C}_6\text{H}_5\text{CH}(\text{OH})\text{CH}_3$
 B 2,4 – DNPH + $\text{C}_6\text{H}_5\text{CH}_2\text{CO}_2\text{H}$
 C hot $\text{NaOH}_{(\text{aq})}$ + $\text{C}_6\text{H}_5\text{CH}_2\text{CHO}$
 D warm acidified $\text{CH}_3\text{CH}_2\text{OH}$ + $\text{C}_6\text{H}_5\text{CONH}_2$

- 27 Which reagent does **not** give fumes of a gas when reacted with the organic compound shown?

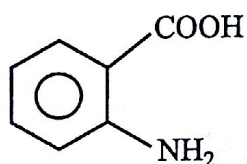


- A bromine liquid
 B ethanoyl chloride
 C hot aqueous sodium hydroxide
 D cold potassium metal
- 28 2-methylbut-2-ene was reacted with hot alkaline KMnO_4 and hot acidified KMnO_4 .

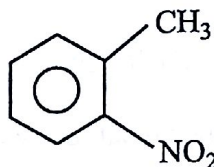
Which row gives the correct products for the reactions?

	hot acidified KMnO_4	hot alkaline KMnO_4
A	CH_3COCH_3 and CH_3COOH	CH_3COCH_3 and CH_3COOH
B	CH_3COCH_3 and CH_3COO^-	CH_3COCH_3 and CH_3COO^-
C	CH_3COCH_3 and CH_3COO^-	$\text{CH}_3\text{COOCH}_3$ and CH_3COOH
D	CH_3COCH_3 and CH_3COOH	CH_3COCH_3 and CH_3COO^-

- 29 Which statement is correct about the compounds?



and



- A They are both soluble in water.
 B They are functional group isomers.
 C The substituent groups on benzene have a positive inductive effect.
 D The substituent groups on benzene have a negative inductive effect.
- 30 In which compound is the carbon-halogen bond hydrolysed most readily by aqueous sodium hydroxide?

- A $\text{CH}_3\text{CH}_2\text{F}$
 B $\text{CH}_3\text{CH}_2\text{Cl}$
 C $\text{CH}_3\text{CH}_2\text{Br}$
 D

Section B

For each of the questions in this section, one or more of the three numbered statements 1 to 3 may be correct.

Decide whether each of the statements is or is not correct. (You may find it helpful to put a tick against the statement(s) which you consider to be correct).

The responses A to D should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

31 Which statement(s) is/are true about Group (II) nitrates?

1. Thermal decomposition decreases down the group.
2. Decomposition temperature increases down the group.
3. Polarisation of nitrate ions by Group (II) cations increase down the group.

32 The bond angle in PH_3 is 97° while that of NH_3 is 107° .

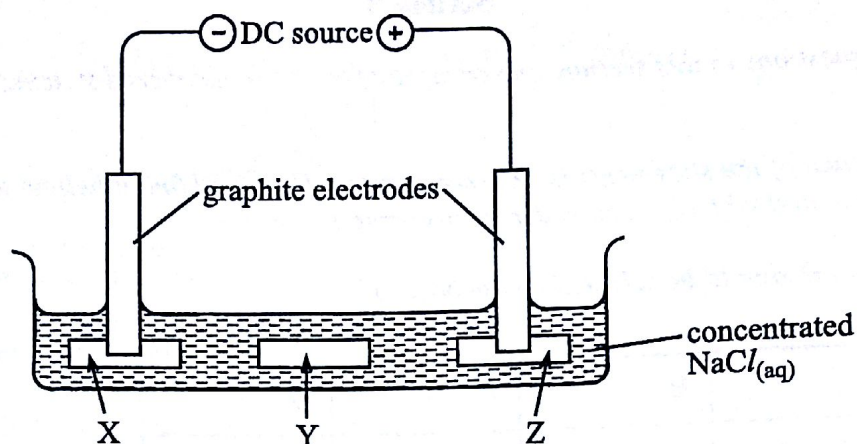
Which statement(s) is/are a correct explanation for this observation?

1. N – H bond length is shorter than P – H bond length.
2. N has greater electronegativity than P.
3. P has a larger atomic radius than N.

33 Which compound(s) would **not** give a pungent smelling gas when treated with concentrated sulphuric acid.

1. NaCl
2. NaBr
3. NaI

- 34 An experiment was set up as shown in the diagram.



X, Y and Z are red litmus papers. After a few minutes, X turned blue, Y remained red and Z turned white.

Which statement(s) is/are correct about these observations?

1. H^+ and Na^+ migrated to the cathode.
2. OH^- and Cl^- migrated to the cathode.
3. H^+ was discharged at the cathode and OH^- became excess.

- 35 Which statement(s) about CaO is/are true?

1. It is used as white wash.
2. It is used as a fertilizer.
3. It is used as a reagent to test for carbon dioxide.

- 36 Which class(es) of reaction may have either an endothermic or exothermic enthalpy change?

1. ionic lattice formation
2. solution formation
3. electron affinity

- 37 Which condition(s) allow the real gas behaviour to approximate ideality?

1. low pressure and large volume
2. low pressure and high temperature
3. high temperature and large volume

38 Which species can act as an electrophile?

1. NO_2^+
2. Cl_2
3. NH_3

39 Which statement(s) is/are true about polythene?

1. The repeat unit is $\left(\begin{array}{c} \text{H} \\ | \\ -\text{C}- \\ | \\ \text{H} \end{array} \right)_n$
2. It has the same empirical formula as ethane.
3. It is non-biodegradable.

40 Which mechanism(s) correctly represent(s) an electrophilic substitution?

